Dual Self-Attention Swin Transformer for Hyperspectral Image Super-Resolution

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***Abstract*— Spatial resolution is a crucial indicator for measuring the quality of hyperspectral imaging (HSI) and obtaining high-resolution (HR) hyperspectral images without any auxiliary information has become increasingly challenging. One promising approach is to use deep-learning (DL) techniques to reconstruct HR hyperspectral images from low-resolution (LR) images, namely super-resolution (SR). While convolutional neural networks are commonly used for hyperspectral image SR (HSI-SR), they often lead to unavoidable performance degradation due to the lack of long-range dependence learning ability. In this article, we propose a dual self-attention Swin transformer SR (DSSTSR) network that utilizes the ability of the shifted windows (Swin) transformer in the spatial representation of both global and local features and learns spectral sequence information from adjacent bands of HSI. Additionally, DSSTSR incorporates an image denoising module using the wavelet transformation method to mitigate the impact of stripe noise on HSI-SR. Our extensive experiments using publicly close-range datasets demonstrate that DSSTSR outperforms other state-of-art HSI-SR methods in terms of three image quality metrics. Furthermore, we applied DSSTSR to the SR of satellite hyperspectral images and achieved improved classification results. Compared to its competitors, DSSTSR exhibits superior performance in enhancing spatial resolution while preserving spectral information. These results suggest that the DSSTSR network has great potential for standardization in remote-sensing image processing and practical applications.**

***Index Terms*— Convolutional neural network (CNN), dual self-attention, hyperspectral image, shifted windows (Swin) transformer, super-resolution (SR).**

I. Introduction

IMAGE super-resolution (SR) is a widely used technique in the field of computer vision to enhance the quality of images. The technique of SR involves transforming low-spatial-resolution images into high-spatial-resolution images, resulting in fine details on edges and textures. As a result, experts in computer vision often apply SR to images in a variety of application fields, including medical image processing [1], [2], [3], security and surveillance [4], [5], and object recognition [6]. As for images collected by optical sensors, spectral resolution also becomes an important factor for measuring image quality. In comparison with three channels (red–green–blue) in natural images, spectral images contain a significantly large amount of detailed information. Hyperspectral images can capture data from hundreds or even thousands of narrow and contiguous wavelengths of the electromagnetic spectrum. This abundant spectral information allows us to identify specific materials or target detection [7], [8] based on their unique spectral signatures. Hyperspectral imaging (HSI) becomes a powerful tool in a broad range of remote-sensing applications such as environmental protection and resource exploration [9], [10]. However, the balance between spatial and spectral resolutions is a dilemma due to the limited energy of light radiation. Remote-sensing images are often taken from space at a very high altitude, resulting in low spatial resolution. Satellite HSI sensors collect hundreds of bands across the wavelength of 400–2500 nm. Current satellites in orbit can only achieve a spatial resolution of 30 m, which is insufficient for many land cover mapping tasks. Hardware improvements are challenging, making it more practical to develop approaches for increasing the spatial resolution of hyperspectral images.

HSI SR often requires auxiliary information. High-resolution (HR) hyperspectral images are commonly generated by fusing with multispectral images (MSIs), panchromatic (PAN) images, or RGB images, but this approach has limitations. One constraint is that HSI and the additional HR images must be simultaneously acquired. The fusion-based SR method requires the co-registration of the auxiliary image and the low-resolution (LR) hyperspectral image. As a result, single hyperspectral image SR (HSI-SR) has gained significant attention in real applications since it does not require additional auxiliary information or geometric registration.

Conventional methods for single SR can be categorized into two types: interpolation and reconstruction. Interpolation involves extrapolating new data points within a range from known data points, such as linear or bicubic interpolation [11]. Reconstruction methods utilize different prior knowledge or constraints for image SR, such as local [12], nonlocal, and sparse priors [13]. However, both categories of methods have drawbacks. Interpolation-based methods often fail to retain

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high-frequency information, causing blurred imaging details, while reconstruction-based methods may not perform robustly on different hyperspectral image datasets.

Recently, deep learning (DL) has emerged as a powerful tool for SR. Its modeling ability is well-suited for establishing the spatial mapping between the LR and HR images. Among various kinds of deep neural networks, convolution neural networks (CNNs) show outstanding performance in SR, particularly for high-dimensional data [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31]. Classic methods include spatial–spectral prior SR (SSPSR) [21], exploring the relationship between 2-D/3-D convolution SR (ERCSR) [18], and the mixed 2-D/3-D convolutional network (MCNet) [26]. These methods and their variants can extract spatial structure information using filtering kernels. However, CNNs often focus too much on local context information and neglect long-range dependency on the data. On the other hand, many CNN-based SR methods ignore the spectral relationship of band images, resulting in the loss of important spectral information during feature extraction. Both global spatial information and spectral context are key to the reconstruction of hyperspectral images.

The transformer network, initially proposed for analyzing sequence data in natural language processing (NLP), provides a good fit for addressing the CNN's limitations. Unlike the CNN, the transformer uses a self-attention mechanism that well captures global information, and it processes the data in a sequential way. However, due to the large number of parameters in the network, the transformer may not be ideal for analyzing high-dimensional data, such as hyperspectral images. Using the original transformer network for HSI-SR may lead to slow training and reduced effectiveness. To address this issue, a shifted window-based shifted windows (Swin) transformer [32] is developed. This approach reduces the number of parameters while still effectively capturing global features, making it ideal for HSI-SR tasks. Another challenge in processing the hyperspectral image is the great noise caused by the imaging sensors. Various types of noise will be generated during the acquisition and transmission of HSI, for example, thermal and quantization noise. Recent studies [33] have shown that noise may increase as image data flow through a deep network. Networks will be more inclined to extract features from noisy textures, leading to erroneous results. Therefore, our study also focused on reducing the noise in HSI to improve the performance of SR.

To resolve the issues mentioned above, we propose a novel transformer-based network called the dual self-attention Swin transformer SR (DSSTSR) network. The key point of DSSTSR mainly includes two modules: one is the image denoising module, and the other is the dual self-attention Swin transformer (DSST) module. The DSST module combines spatial self-attention and spectral self-attention to reduce the distortion of spectral information when extracting spatial information. This design makes the Swin transformer more suitable for HSI-SR tasks. The image denoising module preprocesses the input image, overcomes the problem that the noise will gradually expand in the deep network, and reduces the impact of noise on the SR.

In general, the contributions of this work are summarized as follows.

1) We proposed the DSSTSR network, which initially applied the Swin transformer to HSI-SR. This proposed DSSTSR demonstrated robust performance in HSI-SR at different scale factors.
2) We devised two simple but effective modules in DSSTSR: the image denoising module and the DSST module. The image denoising module used wavelet transformation to reduce the stripe noise. The DSST module performs self-attention feature extraction from spatial and spectral dimensions.
3) Our method outperformed other comparative methods in three image quality metrics, in experiments on CAVE and Harvard datasets. We also conducted experiments on well-acknowledged satellite hyperspectral images (PRISMA, recently launched by the Italian space agency) and demonstrated improved classification of land covers after SR using our proposed method. This has important practical significance, as the proposed method can be widely applied to the processing of satellite images in a standardized manner.

The rest of this article is organized as follows. Section II reviews related work on HSI-SR. The method of DSSTSR is described in Section III. The experimental results are analyzed and discussed in Section IV. Finally, this work is summarized in Section V.

II. Related Work

In this section, this article reviews work on two related aspects: CNN-based methods for single HSI-SR and transformer-based methods.

A. CNN-Based Methods for SR

Different types of convolutional neural networks have been proposed for the SR of a single HSI. This review describes several representative networks.

1) Two-Dimensional-CNN: msiSRCNN [15] was one of the pioneering studies that adopted natural image SR methods for SR of HSI. Early studies (i.e., spectral difference convolutional neural network) [16], [29] directly used 2-D convolutions to extract the spatial information from HSIs, while others attempted to enhance collaboration between LR and HR HSIs by employing collaborative nonnegative matrix factorization (CMNF) and 2-D convolutions [23]. These methods often focused on spatial information, disregarding the importance of spectral information, until the SSPSR [21] network started recognizing the correlation between spectra bands, allowing for better exploration of spectral information.

2) Three-Dimensional-CNN: Three-dimensional full-CNN (3D-FCNN) was proposed to jointly explore spatial and spectral information, resulting in improved SR accuracy and reduced spectral distortion [27]. However, 3-D convolution is computationally expensive, which hampers its application. More efficient 3D-CNN networks, such as the enhanced 3-D convolutional network (E3DN) [30] and 3-D residual in residual dense block (3DRRDB) [31], have since been proposed.

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These models addressed the computational complexity by decomposing the 3-D convolution into continuous spatial and spectral components.

3) Hybrid 2-D and 3-D CNN: The computational complexity of 3D-CNN can be further alleviated by reducing the usage of 3-D convolution [18], [26], [34]. Therefore, a mixed convolution network (MCNet) [26] and its variations (i.e., ERCSR) [18] were proposed. More recently, spectrum and feature context SR (SFCSR), a dual-channel network with 2-D and 3-D convolutions to jointly exploit information from a single band and adjacent bands, was designed [34]. Inspired by the 2-D/3-D correlation between these methods, this article aims to find a better way of using CNN for mining spectral and spatial features and reducing the model's size. This study also explores the combination with the transformer structure to overcome CNN's limitation on capturing global information.

B. Transformer-Based Methods

This section introduces the development of the transformer network and some famous variations.

1) Vision Transformer: The transformer network was first introduced in 2017 for machine translation tasks, and its impressive performance in image processing has led to its popularity in computer vision tasks [35], [36]. Since then, various visual models based on the transformer have been proposed [37], [38], [39], [40]. For example, Chen et al. [37] proposed the image processing transformer (IPT), a transformer-based pretraining model for addressing low-level computer vision tasks such as denoising, SR, and rain removal. Uformer [38] is another popular image restoration method that employs a hierarchical structure with a transformer to effectively capture local and global contextual information. Additionally, for HR image restoration tasks, the restoration transformer (Restormer) network [39] and efficient SR transformer (ESRT) [40], both based on multihead attention, have been introduced.

2) Swin Transformer: As the use of a transformer in language and vision domains are different, a hierarchical transformer utilizing shifted windows, namely the Swin transformer, was developed [32]. This mechanism facilitates capturing long-distance dependencies while reducing the computation and has achieved state-of-the-art results. Swin-based models are now being explored for image restoration [41], [42]. SwinIR [41], one representative SR model, has demonstrated robust performance in reconstructing high-quality images. Further enhancing the efficiency of this method, SwinIR with fast Fourier convolution (SwinFIR) [42] replaces the convolutional layers in SwinIR with the fast Fourier convolution (FFC).

3) Dual Self-Attention: While the single-channel self-attention mechanism cannot fully meet the needs of some complex tasks, researchers have proposed a dual self-attention mechanism [43], [44], [45], [46]. One example is the dual attention vision transformer (DaViT) [43], a transformer-based network composed of spatial and channel self-attention modules. It can effectively extract and classify features with high accuracy. This mechanism has been employed in various applications, such as the dual aspect self-attention-based transformer (DAST) [44] for remaining useful life prediction, the dual self-attention network (DSANet) [46] for multivariate time-series forecasting, and the adaptive dual self-attention network (IDEA-Net) [45] for denoising. Given its powerful capabilities, the dual self-attention mechanism can likely be further enhanced by integrating it with the Swin transformer architecture.

III. Methods

This section presents the DSSTSR network for a single HSI-SR. The network architecture of the DSSTSR is first introduced in Section III-A, and its two key components (i.e., preprocessing and DSST modules) are then elaborated in Sections III-B and III-C.

A. Network Architecture

The DSSTSR network aims to learn an end-to-end mapping function from LR image $\boldsymbol{I}_{\mathrm{LR}} \in \mathbb{R}^{h\times w\times c}$ to the SR image $\boldsymbol{I}_{\mathrm{SR}} \in \mathbb{R}^{rh\times rw\times c}$, where h, w, and c represent the height, width, and spectral channel numbers and r denotes the scale factor. The learning process is mainly based on feature extraction from both spatial and spectral dimensions. The overall architecture of the network is presented in Fig. 1.

1) Preprocessing Module: The preprocessing module contains two parts: band splitting and image denoising [see Fig. 1(a)]. Recent studies [28] and [34] show that the high similarity between adjacent bands has an impact on SR. Therefore, band splitting and reorganizing are included in the preprocessing module. As described in Section I, HSI-SR also requires attention to the influence of noise. Given an LR image $\boldsymbol{I}_{\mathrm{LR}}$, the band set after preprocessing $\hat{\boldsymbol{P}}(i)$ is obtained by

$$\hat{\boldsymbol{P}}(i) = \mathcal{H}_{\mathrm{ID}}(\mathcal{H}_{\mathrm{BS}}(\boldsymbol{I}_{\mathrm{LR}})) \tag{1}$$

where $\mathcal{H}_{\mathrm{BS}}(\cdot)$ is the band splitting process and $\mathcal{H}_{\mathrm{ID}}(\cdot)$ is the image denoising process.

2) Shallow Feature Extraction by CNNs: Convolutions are suitable for extracting shallow features and providing stable optimization for deep feature extraction [47]. Given the denoised image $\hat{\boldsymbol{P}}(i)$, this DSSTSR utilizes a 2-D convolution block, a residual block, and three 3-D convolution blocks [as illustrated in Fig. 1(b)] to extract shallow features from the image. The combination of 2-D and 3-D convolutions enables a better extraction of spatial information while obtaining spectral fidelity. The shallow feature ($\boldsymbol{F}_{\mathrm{SF}}$) can be obtained by

$$\boldsymbol{T} = \mathcal{H}_{\mathrm{Res}}\left(\mathcal{H}_{\mathrm{Conv2}D}\left(\hat{\boldsymbol{P}}(i)\right)\right) \tag{2}$$

$$\boldsymbol{F}_{\mathrm{SF}} = \mathcal{H}_{\mathrm{Fusion}}\left(\mathcal{H}_{\mathrm{Spa}}(\boldsymbol{T}) + \mathcal{H}_{\mathrm{Spec}}(\boldsymbol{T})\right) \tag{3}$$

where $\mathcal{H}_{\mathrm{Conv2}D}(\cdot)$ denotes the 2-D convolution block and $\mathcal{H}_{\mathrm{Res}}$ denotes the residual block. Meanwhile, $\mathcal{H}_{\mathrm{Spa}}(\cdot)$ is a 3-D convolution block with a $1 \times 3 \times 3$ kernel to extract spatial information; $\mathcal{H}_{\mathrm{Spec}}(\cdot)$ denotes another 3-D convolution block with a $3 \times 3 \times 1$ kernel to extract spectral information, and $\mathcal{H}_{\mathrm{Fusion}}(\cdot)$ represents a 3-D convolution block with a $1 \times 3 \times 3$ kernel to fuse the spatial and spectral information.

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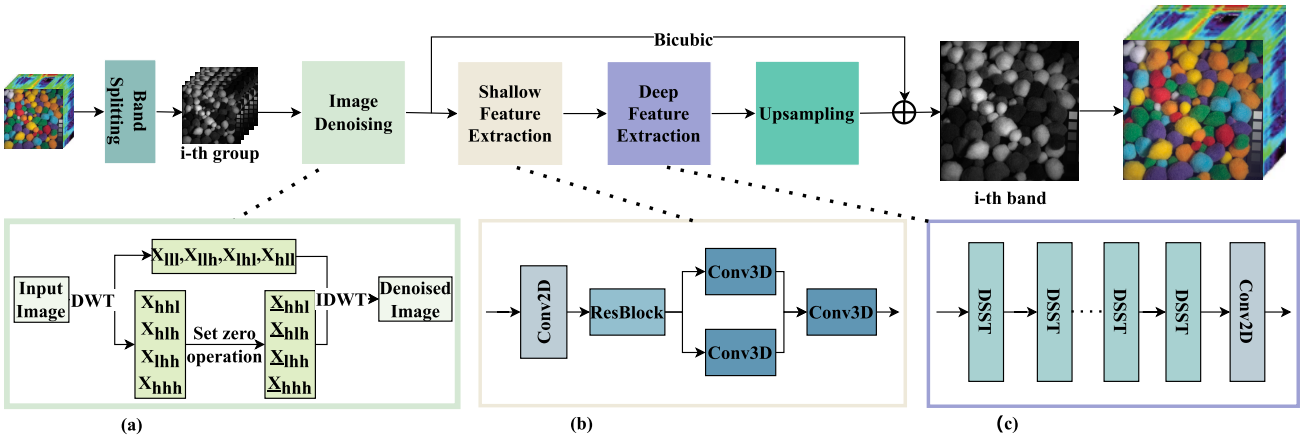

Fig. 1. Architecture of the proposed DSSTSR network. (a) Image denoising module. (b) Shallow feature extraction module. (c) Deep feature extraction module.

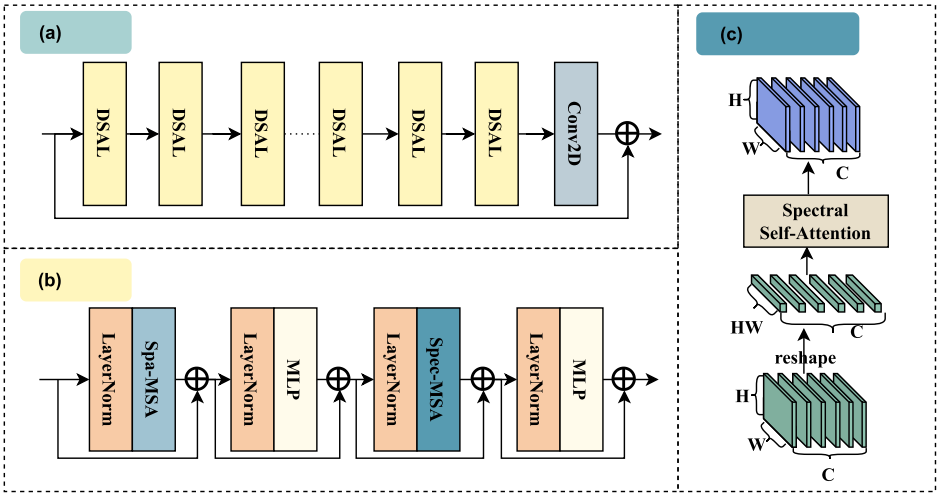

Fig. 2. Modules of the proposed DSSTSR: (a) DSST module; (b) DSAL; and (c) Spec-MSA.

3) Deep Feature Extraction by DSST: The shallow feature $\boldsymbol{F}_{\mathrm{SF}}$ is processed into the deep feature extraction which contains N DSST modules and a convolution layer. The output deep feature ($\boldsymbol{F}_{\mathrm{DF}}$) can be expressed as

$$\boldsymbol{F}_{\mathrm{DF}} = \mathcal{H}_{\mathrm{Conv}}\left(\mathcal{H}_{\mathrm{DSST}_l}\left(\cdots \mathcal{H}_{\mathrm{DSST}_1}(\boldsymbol{F}_{\mathrm{SF}})\cdots\right)\right) \tag{4}$$

where $\mathcal{H}_{\mathrm{DSST}_l}(\cdot)$ denotes the lth DSST module and $\mathcal{H}_{\mathrm{Conv}}(\cdot)$ denotes the convolution layer [the details of the DSST module are shown in Fig. 2(a)–(c)].

4) Image Reconstruction: For image reconstruction, the DSSTSR network upsamples deep features into HR space. In this way, the size of the input and output images can be consistent. This study uses bicubic interpolation as the upsampling method. In the upsampling process, the denoised image and upsampled deep features are put together to obtain the reconstruction output $\boldsymbol{I}_{\mathrm{SR}}$. Adding the original denoised images will make the network focus on extracting high-frequency information, thus reducing the burden of the SR network.

5) Loss Function: Finally, the reconstruction based on DSSTSR is constrained by the L_1 loss function

$$\mathcal{L} = ||\boldsymbol{I}_{\mathrm{SR}} - \boldsymbol{I}_{\mathrm{HR}}||_1 \tag{5}$$

where $\boldsymbol{I}_{\mathrm{SR}}$ represents the SR image obtained through learning, while $\boldsymbol{I}_{\mathrm{HR}}$ represents the original HR image. The symbol $||\cdot||$ refers to the absolute difference between two images, which is calculated pixel-wise. A minimum loss is desirable in order to bring $\boldsymbol{I}_{\mathrm{SR}}$ close to $\boldsymbol{I}_{\mathrm{HR}}$.

B. Preprocessing Module: Band Splitting and Image Denoising

The first part of the network is the preprocessing module that decomposes and denoises the input image with LR ($\boldsymbol{I}_{\mathrm{LR}}$). Considering the high dimensionality and strong spectral correlation of the data, the proposed method adopts the strategy of [28] to provide input with a set of five adjacent bands

$$\boldsymbol{P}(i) = \begin{cases} \left[\boldsymbol{I}^1_{\mathrm{LR}}, \boldsymbol{I}^2_{\mathrm{LR}}, \boldsymbol{I}^3_{\mathrm{LR}}, \boldsymbol{I}^4_{\mathrm{LR}}, \boldsymbol{I}^5_{\mathrm{LR}}\right], & 1 \le i < 3 \\ \left[\boldsymbol{I}^{i-2}_{\mathrm{LR}}, \boldsymbol{I}^{i-1}_{\mathrm{LR}}, \boldsymbol{I}^{i}_{\mathrm{LR}}, \boldsymbol{I}^{i+1}_{\mathrm{LR}}, \boldsymbol{I}^{i+2}_{\mathrm{LR}}\right], & 3 < i < L-3 \\ \left[\boldsymbol{I}^1_{\mathrm{LR}}, \boldsymbol{I}^2_{\mathrm{LR}}, \boldsymbol{I}^3_{\mathrm{LR}}, \boldsymbol{I}^4_{\mathrm{LR}}, \boldsymbol{I}^5_{\mathrm{LR}}\right], & L-3 < i \le 3 \end{cases} \tag{6}$$

where $\boldsymbol{P}(i)$ is the ith band set and L is the total number of bands. Each band set is fed into the network for feature extraction. The final output from the network is a single-band image after SR.

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A wavelet transformation is applied to each band to reduce the image noises since noises propagate through the network attention and pose negative impacts on feature extraction. The wavelet transformation [33] method decomposed high-dimensional data (i.e., spectra in this case) into low- and high-frequency components and reconstructed the original data from the decomposed components through scale and wavelet functions. In this study, noises in the hyperspectral image (mainly high-frequency signals) are removed by wavelet transformation, and the details are presented as follows.

For the image $\boldsymbol{P}(i)$, the image denoising module has two steps. The first step is to decompose the image into eight components through different filters ($\mathcal{R}$)

$$\boldsymbol{X}_{c_0c_1c_2} = \mathcal{H}_{\text{down}\times 2}(\mathcal{R}_{c_0c_1c_2} * \boldsymbol{P}(i)), \quad c_0, c_1, c_2 \in \{l, h\} \tag{7}$$

where $\mathcal{H}_{\text{down}\times 2}$ is a convolution with downsampling, $\mathcal{R}_{lll}$ is a 3-D low-pass (l) filter, and $\mathcal{R}_{llh}, \mathcal{R}_{lhh}, \mathcal{R}_{lhl}, \mathcal{R}_{hll}, \mathcal{R}_{hlh}, \mathcal{R}_{hhl}, \mathcal{R}_{hhh}$ are the 3-D high-pass (h) filters. $\boldsymbol{X}_{lll}$ is the low-frequency information that contains the main texture of the hyperspectral image, and $\boldsymbol{X}_{hll}, \boldsymbol{X}_{lhl}, \boldsymbol{X}_{llh}, \boldsymbol{X}_{hhl}, \boldsymbol{X}_{hlh}, \boldsymbol{X}_{lhh}, \boldsymbol{X}_{hhh}$ are the high-frequency information that delineate the horizontal, vertical, and diagonal details. The noises of the hyperspectral image captured by four high-pass components ($\boldsymbol{X}_{hhl}, \boldsymbol{X}_{hlh}, \boldsymbol{X}_{lhh}, \boldsymbol{X}_{hhh}$) are removed. The remaining high-pass components are kept to prevent too much loss of image information, which can lead to poor feature extraction. The second step is to fuse all these decomposed components for the reconstruction of the original image

$$\hat{\boldsymbol{P}}(i) = \sum_{c_0,c_1,c_2\in\{l,h\}} \mathcal{R}_{c_0c_1c_2} * \mathcal{H}_{\text{up}\times 2}(\boldsymbol{X}_{c_0c_1c_2}) \tag{8}$$

where $\mathcal{H}_{\text{up}\times 2}$ is a convolution with upsampling, $\boldsymbol{X}_{c_0c_1c_2} \in \mathbb{R}^{\lfloor (h/2)\rfloor \times \lfloor (w/2)\rfloor \times c}$, and $\hat{\boldsymbol{P}}(i)$ is the denoised image. Through wavelet transformation, the noisy information has been removed, and its influence on feature extraction can, therefore, be alleviated.

C. DSST Module

The DSST module contains M dual self-attention layers (DSALs) and a 2-D convolution block, as shown in Fig. 2(a). The intermediate features $\boldsymbol{F}_{i,j}$ are iteratively extracted from the shallow features $\boldsymbol{F}_{\text{SF}}$ via DSAL

$$\boldsymbol{F}_{i,j} = \mathcal{H}_{\text{DSAL}}(\boldsymbol{F}_{l,j-1}), \quad j = 1, 2, \ldots, M \tag{9}$$

where $\mathcal{H}_{\text{DSAL}}$ is the jth DSAL in the lth DSST module. Then, the network delivers the DSST output ($\boldsymbol{F}_{i,\text{out}}$) through residual connections

$$\boldsymbol{F}_{i,\text{out}} = \mathcal{H}_{\text{Conv}}(\boldsymbol{F}_{i,M}) + \boldsymbol{F}_{i,0}. \tag{10}$$

The DSAL [see Fig. 2(b)] is used for extracting deep features from hyperspectral images, enhancing the effectiveness of nonlocal long-range dependencies, and reducing spectral distortion.

The DSAL is based on the Swin transformer. The main differences between DSAL and Swin transformer lie in dual self-attention and spectral multihead self-attention (Spec-MSA). The DSAL contains two steps. First, it learns spatial information using the Swin attention in the Swin transformer. During spatial self-attention, the shallow feature $\boldsymbol{F}_{\text{SF}}$ is first split into $(h \times w/m^2)$ nonoverlapping local windows (h: height, w: width, m: window size), and then self-attention is computed for each window through a Swin mechanism. For the local window $\mathcal{S}$, query, key and value are calculated as

$$Q = \mathcal{S}M_Q, \quad K = \mathcal{S}M_K, \quad V = \mathcal{S}M_V \tag{11}$$

where Q, K, V denote the query, key, and value, respectively. The M_Q, M_K, and M_V are the projection matrices. The attention in the window is computed as

$$\text{Attention}(Q, K, V) = \text{SoftMax}(QK^T\sqrt{d} + B)V \tag{12}$$

where B is a learnable relative position code. Second, a spectral self-attention mechanism is incorporated, in order to explore the deep spectral information of hyperspectral images. The input data with size $h \times w \times c$ is reshaped into an image with size $hw \times c$. The linear projections along the spectral dimension ($\hat{Q}, \hat{K}, \hat{V} \in \mathbb{R}^{hw\times c}$) are obtained by

$$\hat{Q} = \mathcal{S}W_Q, \quad \hat{K} = \mathcal{S}W_K, \quad \hat{V} = \mathcal{S}W_V \tag{13}$$

where W_Q, W_K, and $W_V \in \mathbb{R}^{c\times c}$ are the projection matrices across the spectral channel. The self-attention mechanism in the spectral dimension is performed by

$$\text{Attention}\big(\hat{Q}, \hat{K}, \hat{V}\big) = \text{SoftMax}(\sigma \hat{K}^T \hat{Q})\hat{V} \tag{14}$$

where σ is a parameter that reweight the matrix multiplication $\hat{K}, \hat{Q}$.

The DSSTSR network uses a multilayer perceptron (MLP) layer to connect spatial attention and spectral attention. The MLP contains two fully connected layers and one gelu nonlinear layer. It can enhance the feature extraction based on spatial and spectral self-attention mechanisms.

IV. Experimental Results

This section documents experiments of the DSSTSR network on close-range and remote-sensing datasets. All datasets are scaled to the range [0, 1] before feeding into the network. The quality of images after SR is assessed with a peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and spectral angle mapper (SAM). For evaluating the performance of SR on remote-sensing images, and classification is applied to the original image and SR-processed image. For measuring classification performance, criteria of average accuracy (AA), overall accuracy (OA), and kappa coefficient are calculated.

A. Datasets and Implementation Detail

For the evaluation of the proposed DSSTSR, three hyperspectral datasets are used, namely the CAVE [48], the Harvard [49], and PRISMA [50]. The first two datasets are publicly available close-range hyperspectral images, while the latter is a remote-sensing hyperspectral image. The CAVE dataset comprises 32 indoor hyperspectral scenes that are captured using cooled CCD cameras. Each image is of size

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TABLE I

COMPARING DSSTSR WITH THE STATE-OF-THE-ART ALGORITHMS ON THE THREE DATASETS FOR DIFFERENT VALUES OF UPSAMPLING FACTORS. THE TABLE SHOWS THE NUMBER OF PARAMETERS, INFERENCE SPEED, PSNR, SSIM, AND SAM

Dataset	Upscale	Metrics	Bicubic	EDSR	SSPSR	MCNet	ERCSR	**DSSTSR(Ours)**
CAVE	×2	Param($\times 10^6$, M)	–	1.402	19.125	3.009	1.518	3.489
		Inference Speed(s)	1.278	15.987	6.730	77.893	62.647	20.218
		PSNR ↑	41.846	45.598	45.318	47.321	47.442	**48.357**
		SSIM ↑	0.990	0.994	0.994	0.995	0.995	**0.996**
		SAM ↓	2.605	2.531	2.431	2.229	2.144	**2.059**
	×4	Param($\times 10^6$, M)	–	1.549	21.486	3.393	2.069	3.636
		Inference Speed(s)	0.643	14.854	5.238	43.833	45.564	16.747
		PSNR ↑	36.209	39.628	39.765	39.836	39.743	**40.893**
		SSIM ↑	0.964	0.979	0.982	0.982	0.981	**0.984**
		SAM ↓	3.723	3.426	3.194	3.143	3.175	**2.842**
Harvard	×2	Param($\times 10^6$, M)	–	1.402	10.534	3.009	1.516	3.489
		Inference Speed(s)	1.113	15.543	11.616	49.180	37.242	23.235
		PSNR ↑	42.446	45.763	46.141	46.201	46.178	**46.537**
		SSIM ↑	0.967	0.981	0.978	0.982	0.981	**0.983**
		SAM ↓	2.707	2.723	2.639	2.487	2.497	**2.478**
	×4	Param($\times 10^6$, M)	–	1.550	12.894	3.393	2.069	3.636
		Inference Speed(s)	1.136	14.236	9.238	30.888	27.198	17.461
		PSNR ↑	37.884	40.232	40.463	40.718	40.759	**40.983**
		SSIM ↑	0.928	0.942	0.943	0.944	0.945	**0.946**
		SAM ↓	3.279	3.456	3.225	3.201	3.149	**3.141**
PRISMA	×2	Param($\times 10^6$, M)	–	1.556	11.930	3.009	1.516	3.489
		Inference Speed(s)	2.231	23.808	8.988	85.737	42.972	34.346
		PSNR ↑	40.837	41.671	42.716	42.349	43.042	**43.264**
		SSIM ↑	0.952	0.965	0.971	0.968	0.971	**0.972**
		SAM ↓	1.559	1.385	0.962	0.823	0.844	**0.823**
	×4	Param($\times 10^6$, M)	–	1.783	22.212	3.393	2.069	3.636
		Inference Speed(s)	1.357	23.144	7.073	58.865	35.172	24.569
		PSNR ↑	35.912	36.523	36.828	36.512	36.601	**37.841**
		SSIM ↑	0.857	0.891	0.900	0.878	0.881	**0.906**
		SAM ↓	2.167	1.931	1.832	1.438	1.362	**1.218**

TABLE II

ABLATION STUDY ABOUT THE COMPONENTS

Module	Different combinations of modules			
Image Denoising module	✘	✔	✘	✔
DSST module	✘	✘	✔	✔
PSNR	46.9100	47.1300	47.6180	48.3570
SSIM	0.9957	0.9959	0.9959	0.9962
SAM	2.1150	2.0860	2.0800	2.0590

512 × 512 and contains 31 spectral bands ranging from 400 to 700 nm. For the training, validation, and testing sets, we randomly select 20, 6, and 6 images, respectively. The Harvard dataset includes 50 indoor and outdoor scenes captured by Nuance FX, CRI Inc. cameras, with a wavelength range of 420–720 nm, and the size of each image is 1040 × 1392. We randomly select 40 images as the training set, 5 as the validation set, and 5 as the testing set. Lastly, the PRISMA (PRecursore IperSpettrale della Missione Operativa) [50] dataset is captured by a hyperspectral satellite launched by the Italian Space Agency. The satellite is placed in a sun-synchronous orbit at an altitude of 700 km. This imaging system captures images with a spatial resolution of 30 m and a spectral range of 400–2500 nm, consisting of 233 spectral bands with a size of 1098 × 1098 pixels. To train the network, the image is divided into three sets: a training set consisting of 80% of the regions from the image, a validation set of 10%, and a test set of 10%. LR hyperspectral images for training are generated by bicubic downsampling and introduce a 5 × 5 blur kernel to enhance the robustness of the process. The original image is used as the reference for evaluating the SR of HSI.

This study compares the performance of the proposed network with other similar methods, including the bicubic interpolation method, EDSR [51], SSPSR [21], MCNet [26], and ERCSR [18]. It is worth noting that the EDSR [51] was initially intended for the SR of natural images, whereas the remaining methods are developed for the analysis of hyperspectral images. Therefore, the input and output sizes of the EDSR network are adapted to suit the spectral dimension of HSI. The experimental analysis involves two scale factors of 2× and 4×, and the hyperparameters of each method are optimized to obtain the best possible performance. The comparison results of the three datasets, including PSNR, SSIM, and SAM, number of parameters, and inference speed are presented in Table I. From the table, the DSSTSR generally exhibits better performance compared to its competitors in terms of PSNR, SSIM, and SAM. The detailed comparisons are presented in Sections IV-C and IV-D.

In this study, we adopt the L_1 loss function as the constraint and define the number of filters as 64. We utilize the optimizer Adam ($\beta_1 = 0.9$ and $\beta_2 = 0.999$) to train the network, by setting training epochs as 175. We initialize the learning rate of all layers to 10^{-4}, which is halved by every 35 epochs. All experiments are implemented using the Pytorch framework and accelerated by NVIDIA A40 GPU.

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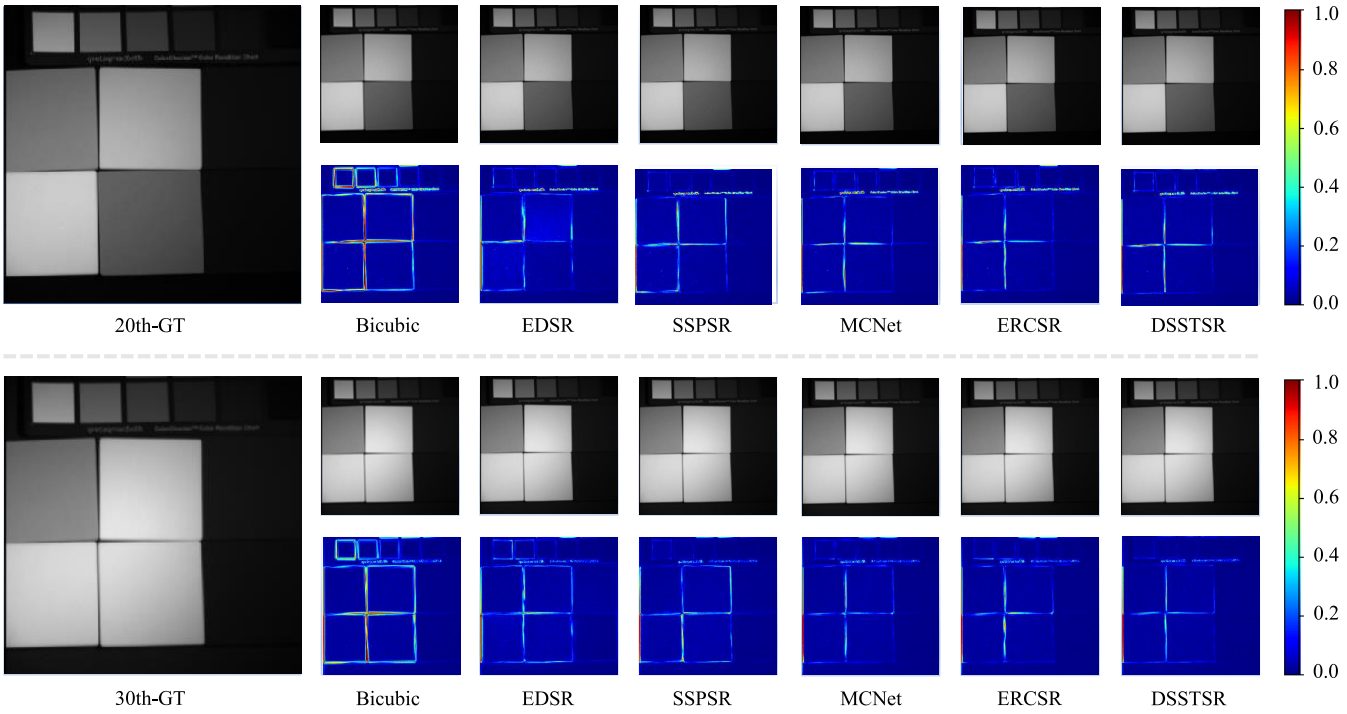

Fig. 3. Absolute error map comparisons for the image sponges_ms at 600 nm (20th band) and 690 nm (30th band) on the CAVE dataset.

B. Ablation Study

To investigate the impact of the image denoising module and DSST module, we conduct ablation experiments on the CAVE dataset. The results in Table II review that the classic Swin transformer without the image denoising module and DSST module produce the worst performance, with PSNR of 46.9100, SSIM of 0.9957, and SAM of 2.1150. This highlights the positive impact of the two modules on the reconstruction performance. The incorporation of either module has led to better indicators. The addition of the image denoising module improves the PSNR by 0.2200, the SSIM by 0.0002, and reduces the SAM by 0.0290 compared to the classic Swin transformer framework. Similarly, after adding the DSST module, the PSNR increases by 0.7080, the SSIM increases by 0.0002, and the SAM decreases by 0.0350. Notably, the combination of the two modules increases the PSNR by more than 1.3000 dB, improves the SSIM by 0.0005, and decreases the SAM by 0.056.

An important trend from Table II is that the combination of the Image Denoising module and DSST module effectively mitigated the spectral loss in the reconstructed image. This can be attributed to the removal of noise and the improved preservation of spectral information. These improvements are evidenced by higher values of PSNR and SSIM, as well as lower SAM values.

C. Experimental Results on CAVE and Harvard Datasets

The proposed DSSTSR is tested on two public datasets, CAVE and Harvard, respectively. Table I presents all evaluation metrics for six methods. For the metrics of PSNR, SSIM, and SAM, the proposed DSSTSR outperforms other networks in all evaluation metrics. Specifically, the average PSNR of DSSTSR surpasses that of the ERCSR method by more than 0.915 dB at $2\times$ scale and 1.056 dB at $4\times$ scale on the CAVE dataset, and by 0.336 dB at $2\times$ scale and 0.225 dB at $4\times$ scale on the Harvard dataset. The EDSR [51] network for natural image SR achieves high spatial fidelity (PSNR $=$ 45.598 dB), however, it suffers from severe spectral distortion (SAM $=$ 2.531). By contrast, methods that jointly learn spatial and spectral information, such as SSPSR [21], MCNet [26], and ERCSR [18], achieve better spectral preservation. Our DSSTSR network achieves high spatial fidelity and delivers the greatest spectral similarity. This result may be attributed to the mechanism of the Swin transformer inherited in the DSST module. The DSST module is capable of processing both spatial and spectral information, resulting in better feature extraction. The improvement can also be ascribed to noise reduction before feature extraction.

Figs. 3 and 4 display the absolute error maps for bands 20 and 30 on the CAVE and Harvard datasets, respectively. In both figures, the images reconstructed by DSSTSR deliver smaller errors. For the CAVE dataset (see Fig. 3), our proposed DSSTSR achieves the best spatial fidelity in restoring the original image's spatial information. Moreover, our method exhibits excellent performance in reconstructing edges and font structures. The same spatial fidelity is also conveyed in experiments on the Harvard dataset (see Fig. 4). Since DSSTSR performs well in reconstructing tree branch details and window frames. This is probably because DSSTSR reduces the impact of imaging noise in deep networks leading to higher-fidelity SR images.

Regarding the reconstructed spectral reflectance of the two datasets, Figs. 5 and 6 illustrate that the spectral curve

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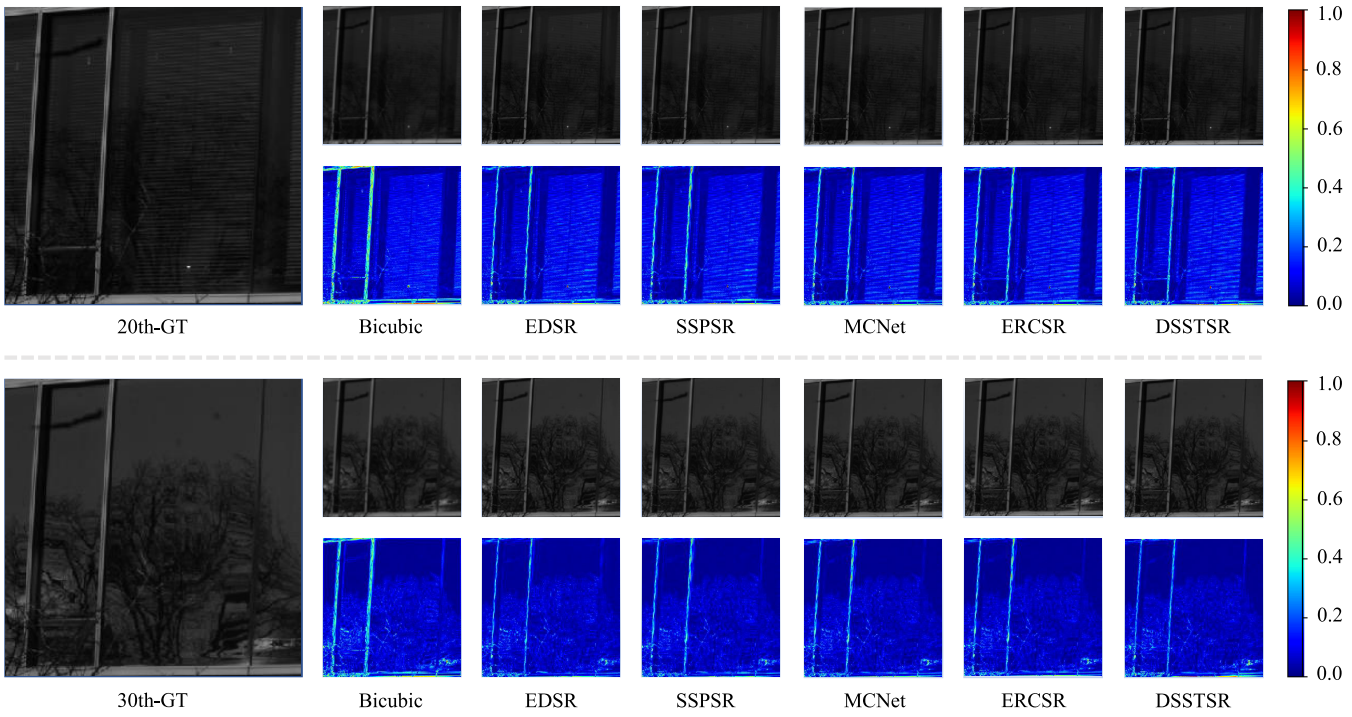

Fig. 4. Absolute error map comparisons for the image imgf3 at 620 nm (20th band) and 710 nm (30th band) on the Harvard dataset.

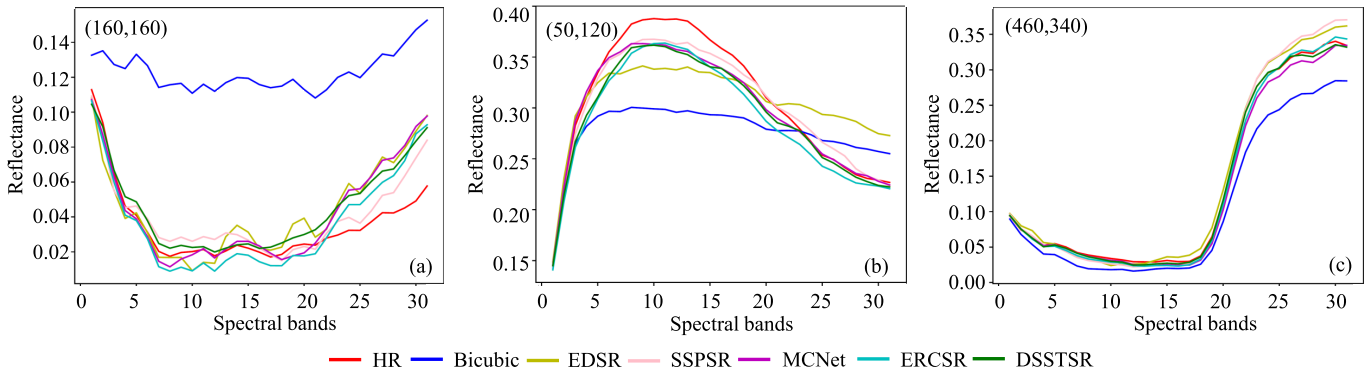

Fig. 5. Visual comparison of spectral distortion for the image sponges_ms at three pixel positions on the CAVE dataset: (a) spectrum of pixel position (160, 160); (b) spectrum of pixel position (50, 120); and (c) spectrum of pixel position (460, 340).

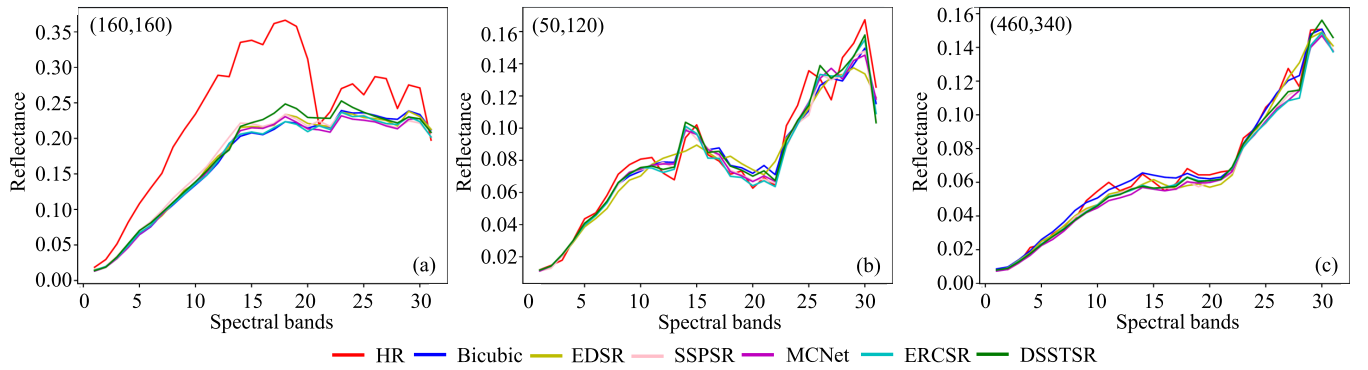

Fig. 6. Visual comparison of spectral distortion for the image imgf3 at three pixel positions on the Harvard dataset: (a) spectrum of pixel position (160, 410); (b) spectrum of pixel position (190, 190); and (c) spectrum of pixel position (230, 230).

generated by the proposed DSSTSR is closer to the ground truth (GT) or slightly weaker in certain bands, compared to other methods. The DSSTSR shows similar or slightly worse performance than the suboptimal method (ERCSR). This might be owing to the band grouping that learns spectral information in the preprocessing module, or the feature extraction in the DSST module that normalizes the spatial–spectral consistency of images.

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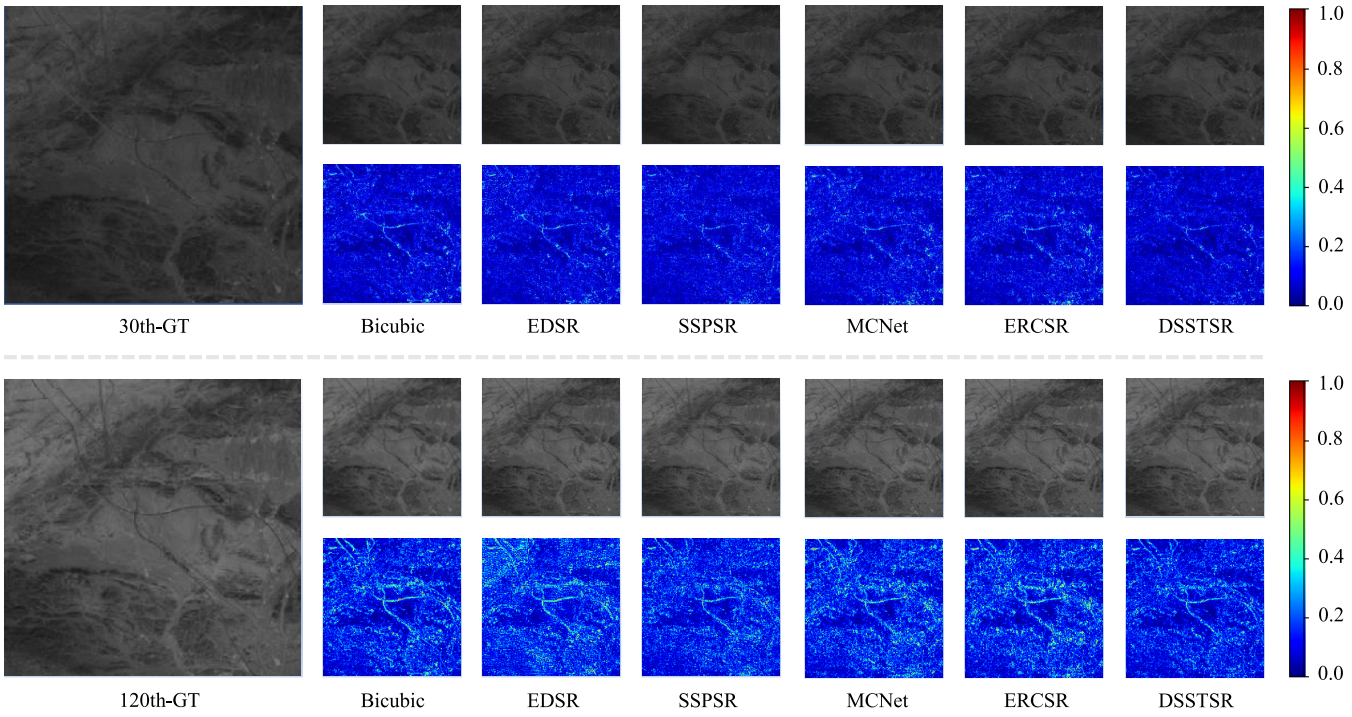

Fig. 7. Absolute error map comparisons for the image at 30th and 120th band on the PRISMA dataset.

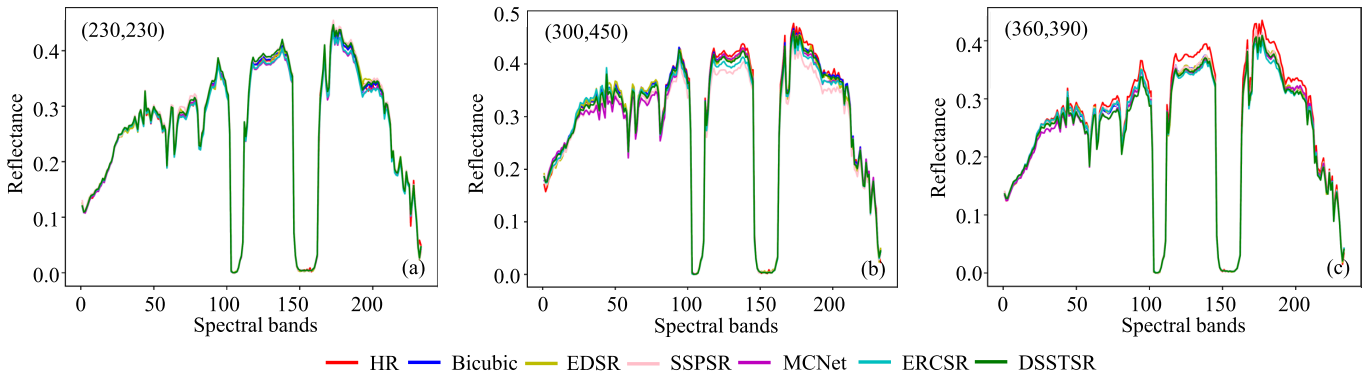

Fig. 8. Visual comparison of spectral distortion at three pixel positions on the PRISMA dataset: (a) spectrum of pixel position (230, 230); (b) spectrum of pixel position (300, 450); and (c) spectrum of pixel position (360, 390).

D. Experimental Result on the PRISMA Dataset

We also conduct experiments on the PRISMA remote-sensing image dataset to demonstrate the practicability of DSSTSR. Table I presents the five metrics (inference speed, number of parameters, PSNR, SSIM, and SAM) of all methods, and DSSTSR outperforms the other methods for both scales. Compared to the ERCSR method, DSSTSR improves PSNR by 0.222 dB and SSIM by 0.001 at 2× scale. Compared to the SSPSR method, DSSTSR improves PSNR by 1.031 dB, increases SSIM by 0.006, and reduces SAM by 0.144 at 4× scale. In comparison to the EDSR method, the DSSTSR decreases SAM values (from 1.931 to 1.218), indicating a high level of spectral consistency. The improvements are more evident when the scale factor is 4. The improvement in PRISMA is not as significant as in the case of the CAVE and Harvard datasets. This is plausible since the SR of the PRISMA dataset has a small training set. The small number of training sets highlights the robustness of DSSTSR and indicates its ability to handle real-case scenarios. SR in practical applications would require a high-level spatial–spectral consistency and a subtle spectral distortion, which has been achieved through the dual self-attention mechanism in DSSTSR.

Fig. 7 visualizes the absolute error map of the reconstructed hyperspectral image versus the GT for the PRISMA dataset at a scale factor of 4. The absolute error map produced by DSSTSR displays darker colors in most areas, indicating a high level of spatial consistency between the original hyperspectral image and the reconstructed image. From both the 30th and 120th grayscale images in Fig. 7, the results produced by bicubic and EDSR are relatively blurred, the ones produced by SSPSR, MCNet, and ERCSR methods show a certain amount of noises. The DSSTSR generates an absolute error map with fewer errors, particularly in road areas, more clearly

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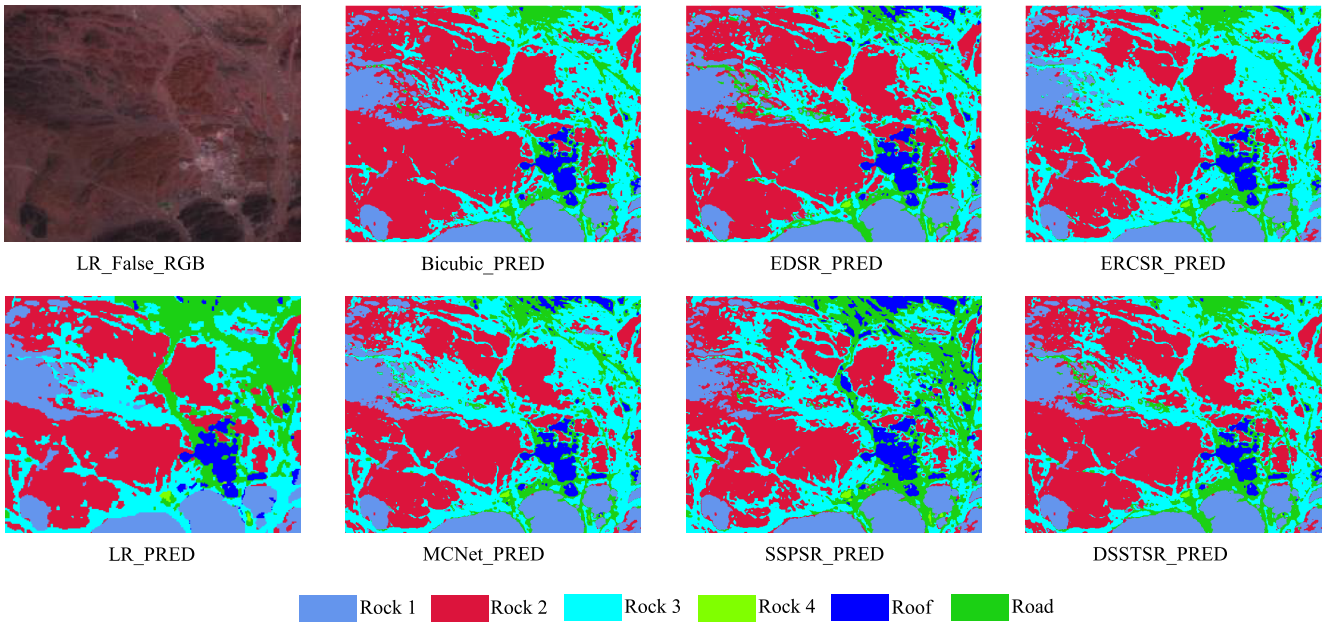

Fig. 9. LR and SR image classification prediction maps.

TABLE III
CLASSIFICATION EXPERIMENTAL RESULTS OF DSSTSR AND FIVE COMPARISON METHODS ON THE PRISMA DATASET

	LR	Bicubic	EDSR	ERCSR	MCNet	SSPSR	DSSTSR(Ours)
AA	90.36%	92.75%	94.73%	95.11%	95.32%	94.94%	**96.15%**
OA	84.50%	89.31%	90.82%	91.39%	91.65%	91.34%	**92.90%**
Kappa	80.96%	86.79%	88.67%	89.35%	89.66%	89.29%	**91.92%**

defined edges and reduced noises. In general, DSSTSR enables better road reconstruction, spatial continuity, and consistency. Fig. 8 shows the spectral curves after SR of the PRISMA dataset, in which DSSTSR performs well in reconstructing the first 100 bands but slightly worse than the MCNet method in the bands 100–150.

To demonstrate the impact of SR on land cover classification, the classification experiment is conducted on the PRISMA dataset after 2× SR. To train the classification, six land cover types (four rocks, one road, and one roof) are defined based on expert knowledge, and the CNN-HSI [52] classifier is employed. For each class, 30 sample points are randomly selected for training, and experiments are conducted ten times in order to obtain an average result. DSSTSR produces the best outcome, with an AA of 92.90%, OA of 96.15%, and Kappa of 91.92%, among all methods. Compared to the classification based on the original image, all SR methods achieve improved performance on SR images at 2× scale (see Table III). Among these, the increase made by DSSTSR is the greatest, for example, it improves AA by 0.83%, OA by 1.30%, and Kappa by 2.26% in comparison with the MCNet method. Furthermore, the improved classification is demonstrated through visualizing prediction maps from all methods in Fig. 9. The classification prediction maps based on SR images have richer edge details than the original image. The DSSTSR correctly classifies rocks (light green in the upper right corner), which are mistakenly predicted as roofs (blue) by other methods. The ERCSR does not address this mistake, although it exhibits good performance in SR tasks as presented above. For the classification of roads in the lower right corner, our method produces clearer boundaries. The increase in spatial resolution from 30 to 15 m would bring great benefits to remote-sensing tasks based on PRISMA datasets. This can obtain good-quality images with higher spatial resolution, facilitating practical applications that aim to classify small targets.

V. CONCLUSION

In this article, we have proposed a novel deep neural network for a single HSI-SR. The proposed method, called DSSTSR, utilizes a DSST module to overcome the limitations of CNN-based and classic transformer-based techniques. It leverages the dual self-attention mechanism to capture the long-range dependencies and generate high-quality hyperspectral images with enhanced spatial resolution and reduced noises. The effectiveness of DSSTSR was demonstrated through multiple experiments on benchmark datasets and satellite images, on which it outperformed state-of-the-art methods in both quantitative and qualitative evaluations. The images derived from DSSTSR exhibited sharper edges and more texture details, indicating superior spatial fidelity. Additionally, the increased classification accuracy achieved through the use of DSSTSR on the PRISMA satellite image emphasizes its potential for numerous remote-sensing applications. Future work will focus on improving the extraction of texture information and its application in real-world scenarios.

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